

Supporting Evidence & Key References

How existing standards, data, and legislation ground the NewsTrace proposal

1

C2PA — Coalition for Content Provenance and Authenticity

What it is: An open industry standard developed by Adobe, BBC, Microsoft, Google, and others for cryptographically attaching provenance metadata to digital media content.

How it supports NewsTrace: NewsTrace extends C2PA's provenance logic — currently applied to images and video — to text-based journalism, filling a documented gap in the standard. The SHA-256 hashing and append-only log in NewsTrace follow the same trust model.

<https://c2pa.org> · *Supported by 100+ organizations worldwide*

2

W3C PROV — Provenance Data Model (World Wide Web Consortium)

What it is: A W3C specification providing a formal, standardized model for expressing provenance on the web. Defines three core concepts: entities (what was produced), activities (how it was produced), and agents (who produced it).

How it supports NewsTrace: The NewsTrace Provenance Recorder maps directly onto this model — every text contribution is an entity, the writing session is the activity, and the journalist or AI tool is the agent. Using W3C PROV ensures interoperability with any system that reads this standard.

<https://www.w3.org/TR/prov-overview/> · *W3C Recommendation*

3

EU Artificial Intelligence Act (2024)

What it is: **Landmark** European legislation requiring that AI-generated content be disclosed to the public. Came into force in 2024 and sets a global precedent for AI transparency obligations across sectors including media.

How it supports NewsTrace: NewsTrace provides the technical implementation layer that the EU AI Act demands but does not specify. The Transparency Label directly fulfils Article 50's disclosure requirement for AI-generated content, while the Provenance Log provides the audit trail regulators need for enforcement.

Official text: eur-lex.europa.eu · *Applies across all EU member states*

4

Reuters Institute Digital News Report (2024)

What it is: Annual global survey of news consumption and trust published by the Reuters Institute for the Study of Journalism at Oxford University. Tracks attitudes toward AI in journalism across 47 countries.

How it supports NewsTrace: The 2024 report documents that over 60% of news organizations are already using AI in their content workflow, yet consistent public disclosure remains absent. This data directly establishes the urgency and scale of the problem NewsTrace addresses.

reutersinstitute.politics.ox.ac.uk/digital-news-report/2024

5

Content Credentials — contentcredentials.org

What it is: The public-facing implementation of C2PA, led by Adobe. This allows creators to attach verified provenance information to images and videos, which platforms like LinkedIn and X display to users as a trust signal.

How it supports NewsTrace: Demonstrates that provenance labelling at scale is technically feasible and publicly adoptable. NewsTrace is the equivalent system for text journalism — the same concept, applied to the written word.

contentcredentials.org · *Already deployed by major platforms*

6**Clerwall, C. (2014) — 'Enter the Robot Journalist', Journalism Practice**

What it is: One of the first peer-reviewed studies on reader perceptions of AI-written news. Found that readers could not reliably distinguish AI-generated from human-written sports reports, raising early questions about transparency and trust.

How it supports NewsTrace: Provides the academic foundation for the reader-facing component of NewsTrace. If readers cannot distinguish AI from human writing, a clear, standardized transparency label is not optional — it is essential.

doi.org/10.1080/17512786.2014.883116 · *Cited 400+ times*

The gap NewsTrace fills

References 1 and 5 show that provenance infrastructure works for images and video. Reference 3 shows that law now requires it for all AI content. References 4 and 6 show that readers need it urgently. Reference 2 provides the formal technical model to build it on.

No existing tool combines all of these into an open-source, text-specific, journalism-ready system. That is the gap NewsTrace proposes to fill.